

Comparison of a Pseudocontinuous, Heterogeneous 2D Conductive Monolith Reactor Model to a 3D Computational Fluid Dynamics Model

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ABSTRACT: Conductive catalytic honeycombs of a low-void fraction have gained renewed interest for use in the production of bulk chemicals because of favorable heat transfer properties and a low pressure drop compared to fixed-bed reactors. In this work, a pseudocontinuous, heterogeneous 2D conductive honeycomb reactor model is compared to a detailed 3D computational fluid dynamics model for the case of an irreversible, exothermic first order reaction with emphasis on the description of heat transfer. Excellent agreement in terms of maximum temperature and conversion is found for moderate conditions preferable for technical purposes when using the symmetric model for calculation of the effective radial heat conductivity. Deviations of maximum temperatures at harsher conditions are attributed to the use of global heat and mass transfer coefficients in the 1D channel model and the inherent assumption of a radially fully segregated flow in the continuum approach.

1. INTRODUCTION

Monolith catalysts are nowadays widely employed in catalytic combustion¹ and exhaust gas cleaning, both for mobile² and stationary³ applications. During the last years they have aroused interest also for use in the production of bulk chemicals since in addition to the low pressure drop in comparison to conventional pellet filled fixed-bed reactors the continuous solid matrix of honeycombs might offer beneficial heat transport properties. The heat transport in fixed-bed reactors for heterogeneously catalyzed gas phase reactions is often limited by the radial heat dispersion, which is dominated by convective heat transport at high particle Reynolds numbers. However, to exploit beneficial heat transfer properties of honeycomb catalysts, different structures and materials are needed in comparison to that needed for honeycombs being used for exhaust gas applications.⁴ In these cases mainly ceramic honeycombs of intrinsic low thermal conductivity are used in adiabatic operation. Therefore, for example metallic Al⁵ and Cu⁶ honeycombs have been developed to improve the heat transfer properties by shifting the main transport mechanism from convection to conduction in a continuous solid phase. A proof of principle was accomplished by Groppi et al.⁷ They studied the partial oxidation of *o*-xylene to phthalic anhydride up to the pilot scale using a tube reactor filled with catalyst-coated Al honeycombs. In comparison to a reactor filled with pellets a higher temperature level proved advantageous, which, however, could be achieved with lower axial temperature differences, thus ensuring safe operation and allowing equal or even better production rates than the fixed-bed reactor. Besides partial oxidations, conductive honeycombs have also been proposed for other exothermic processes including Fischer–Tropsch⁸ and synthetic natural gas⁹ synthesis.

Concerning the modeling of conductive honeycomb catalysts for polytropic operation, mainly continuum models of different complexity are employed. A pioneering continuum model was proposed by Groppi and Tronconi¹⁰ in 1996 in close analogy

to fixed-bed reactor models. The model was then validated by a state-of-the art discrete model. In this discrete model, 1D gas phase mass balances and solid phase energy balances are formulated for individual channels after reconfiguration to concentric rings. A differential-algebraic system is finally set up by coupling the balances of channels representative for the different rings via a finite difference scheme that is directly related to the reconfigured monolith structure.

The key parameter in continuum models is the effective radial heat dispersion coefficient which is calculated according to semiempirical thermal resistance models. The series model proposed by Groppi and Tronconi,¹⁰ has also been extended by taking the catalyst layer explicitly into account, and it has been compared to a so-called parallel model as well as the numerical finite element method (FEM) solution of an isolated channel in stagnant fluid exposed to a constant heat flux.¹¹ It was found that the exact solution is in between the prediction of the series and parallel model for a large variety of void fractions. Similar thermal resistance models have also been applied successfully for diesel particulate filters.¹² Recently, an improved model for the calculation of the effective thermal dispersion has been published.¹³ Also on the basis of an FEM simulation of an imposed heat flux perpendicular to a single channel, this symmetric thermal resistance model proved best compared to the other resistance models and results were very close to the exact solution.

In this research note we present a comparison of a pseudocontinuous, heterogeneous 2D conductive monolith reactor model using the symmetric thermal resistance model to a 3D computational fluid dynamics (CFD) model where fluid flow as well as heat and mass transport are coupled in the

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presence of an exothermic chemical reaction. It has previously been shown that the effective thermal conductivity models are also capable to describe heat transfer properties of a circular arrangement of channels for a coaxial heat source under nonreactive conditions and stagnant fluid.¹¹ Also 1D, 2D, and 3D single channel models have been compared for adiabatic operation (see e.g., the review of Chen et al.).¹⁴ Generally, it was reported that 1D models do not suffice for an adequate description under all operation conditions.

To the best of our knowledge, a comparison to a 3D CFD model under reactive conditions at polytropic operation comprising channel and tube scale has not been presented yet. This might, however, be preferable for validation since it takes the local heat release at the metallic wall into account, which differs from the boundary conditions imposed for the derivation of the effective thermal conductivity models. Also it enables a critical discussion of the continuum approach for low channel number monoliths applied in practice for highly exo- or endothermic reactions, in which the scale of a channel diameter differs only little to the macroscale, that is, the monolith diameter.

2. MODEL DEVELOPMENT

2.1. Reference Case. For the comparison of the models, an exothermic, irreversible first-order reaction is studied. The geometry of the honeycombs is similar to those employed by Groppi et al.⁷ for the *o*-xylene oxidation. The parameters of the reference case are summarized in Table 1. If not stated

Table 1. Parameters and Conditions for the Reference Case ($DaI = 2.0$)

d_{channel}	4 mm
d_{tube}	2.5 cm
m	5 mm
L	25 cm
δ	100 μm
W_t	0.376 g/s
D_A	$2 \cdot 10^{-5} \text{ m}^2/\text{s}$
c_p	1 kJ/(kg·K)
λ_g	0.05 W/(m·K)
λ_s	30 W/(m·K)
M	100 g/mol
k_0	$1.62 \cdot 10^9 \text{ s}^{-1}$
E_A	80 kJ/mol
ΔH_R	−1100 kJ/mol
μ	$2.5 \cdot 10^{-5} \text{ Pa}\cdot\text{s}$
$w_{A,\text{feed}}$	0.05
T_{feed}	520 K
p_{tot}	3.18 bar

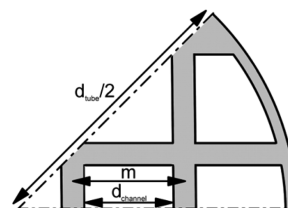
otherwise, all results relate to these conditions. We chose a rather low value of the thermal conductivity of the solid material ($\lambda_s = 30 \text{ W m}^{-1} \text{ K}^{-1}$) in order to be able to compare the models also at sharp temperature profiles which is more demanding in contrast to flat profiles resulting at optimized honeycomb structures made of Al or Cu. The assumptions for the comparison are as follows:

(1) Diffusion limitations inside the washcoat are not considered, since the focus is laid on the heat transfer. The incorporation of a Thiele modulus approach for calculation of effectiveness factors is straightforward but would increase the complexity of the comparison.

(2) The temperature of the lateral surface is set constant and equal to the inlet temperature. The issue of the resistance between honeycomb and tube wall, the so-called gap resistance, is not addressed here. Nevertheless, the gap resistance could be implemented in both models via a heat transfer coefficient for the lateral surface and would not cause further differences between the models.

(3) The flow is distributed equally over all channels though the pressure drop might slightly differ due to different hydraulic diameters of the channels (see Scheme 1).

Scheme 1. Honeycomb Geometry



(4) For the heat transfer, the resistance caused by the washcoat is not taken into account. The conductivity of the solid phase is assumed as uniform.

(5) Material and gas phase properties are assumed independent of pressure and temperature.

(6) The first 5 cm of the honeycomb are assumed to be catalytically inactive. Under harsh conditions, maximum reaction rates and solid phase temperatures are observed in the inlet of the reactor, and the results might be influenced by inlet effects.

(7) The washcoat is distributed uniformly over the metallic walls; that is, the washcoat thickness is constant for the 3D model and the washcoat fraction is not a function of space for the continuum model, respectively.

(8) Heat transfer via radiation is neglected.

2.2. Continuum Model. The pseudocontinuous, heterogeneous 2D monolith reactor model is adapted from Groppi and Tronconi.⁴ It is based on a 1D description concerning the single channel level, which means that global heat and mass transfer coefficients are employed for describing heat and mass transfer to the washcoat layer. Since emphasis is laid on heat transfer properties the model is simplified by assuming constant pressure. The effective radial heat dispersion coefficient is calculated according to the symmetric model proposed in ref 13. As is the model by Groppi and Tronconi,⁴ axial dispersion of mass is neglected. The compartment in individual channels prohibits a radial dispersion of mass. Stationary mass and enthalpy balances for the solid and the gas phase are formulated as follows:

Gas phase

mass balance

$$W \frac{\partial w_{A,g}}{\partial z} + \frac{4d}{m^2} K_m (w_{A,g} - w_{A,s}) = 0$$

enthalpy balance

$$W c_p \frac{\partial T_g}{\partial z} + \frac{4d}{m^2} K_h (T_g - T_s) = 0$$

Solid phase

mass balance

$$\frac{4d}{m^2} \frac{K_m}{\rho_g} (w_{A,g} - w_{A,s}) - \xi \frac{T_g}{T_s} k w_{A,s} = 0$$

enthalpy balance

$$\lambda_{e,ax} \frac{\partial^2 T_s}{\partial z^2} + \lambda_{e,r} \left(\frac{\partial^2 T_s}{\partial r^2} + \frac{1}{r} \frac{\partial T_s}{\partial r} \right) + \frac{4d}{m^2} K_h (T_g - T_s) + \xi (-\Delta H_r) \rho_g \frac{T_g}{T_s} k \frac{w_{A,s}}{M_A} = 0$$

Boundary conditions

$$z = 0: \quad \frac{\partial T_s}{\partial z} = 0 \quad T_g = T_w \quad w_{A,g} = w_{A,feed}$$

$$z = L: \quad \frac{\partial T_s}{\partial z} = 0$$

$$r = 0: \quad \frac{\partial T_s}{\partial r} = 0$$

$$r = R: \quad T_s = T_w$$

2.3. 3D CFD Model. Owing to the symmetry of the honeycomb, only $1/8$ is incorporated in the model (see Scheme 1). Symmetric boundary conditions are set to the cut faces. The fluid flow inside the channels is modeled as laminar, accordingly no slip boundary conditions are imposed. Mass transfer is simplified according to Fickian diffusion.

Gas phase

mass balance

$$\rho(\mathbf{u} \cdot \nabla) w_A = \nabla \cdot (\rho D_A \nabla w_A)$$

enthalpy balance

$$\rho c_p (\mathbf{u} \cdot \nabla T) = \nabla \cdot (\lambda \nabla T)$$

momentum balance

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot \left(-p \mathbf{I} + \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) - \frac{2}{3} \mu (\nabla \cdot \mathbf{u}) \mathbf{I} \right)$$

continuity

$$\nabla \cdot (\rho \mathbf{u}) = 0$$

Solid phase

enthalpy balance

$$0 = \nabla \cdot (\lambda \nabla T)$$

Boundary conditions:

Reacting boundary

mass flux

$$j_A = k_0 \cdot \exp\left(-\frac{E_A}{RT}\right) c_A M_A \delta$$

heat flux

$$h = k_0 \cdot \exp\left(-\frac{E_A}{RT}\right) c_A \delta (-\Delta H_R)$$

no slip (volume conserving reaction)

$$\mathbf{u} = 0$$

lateral surface

$$T = T_w$$

Inlet

gas phase

$$w_A = w_{A,feed} \quad T = T_w \quad u = u_{feed}$$

solid phase

$$\mathbf{n} \cdot \nabla T = 0$$

Outlet

$$\mathbf{n} \cdot \nabla w_A = 0 \quad \mathbf{n} \cdot \nabla T = 0 \quad p = p_{tot}$$

Both models are solved in Comsol Multiphysics V4.3. Mass fractions and temperatures are discretized with a second-order discretization scheme in the continuum model. The fully coupled linear system is solved with the MUMPS direct solver. In the 3D model, the pressure and velocity fields are discretized with a first-order, and temperature and mass fractions are discretized with a second-order scheme. The iterative solver GMRES is used for solving the linear system.

3. RESULTS AND DISCUSSION

3.1. Isothermal Conditions. To verify the implementation of both models, results for an isothermal, first-order reaction have been compared to the analytical solution of an ideal plug flow reactor (PFR). The diffusion coefficient had to be increased compared to the reference conditions to ensure a fully kinetic regime not affected by mass transfer to the wall surface. As illustrated in Figure 1, the conversion in the

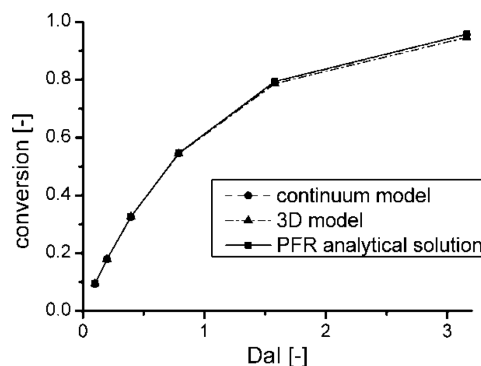


Figure 1. Validation at isothermal operation.

continuum model is identical to the analytical solution while the 3D model slightly underestimates the results at high Damköhler numbers. It has been checked that this is not caused by a mass transfer limitation. A further reason for the discrepancy might be the nonideality of the residence time distribution compared to plug flow. As the cumulative residence time distribution in Figure 2 shows, the flow behavior in the continuum model resembles plug flow since no axial dispersion term is considered in the gas phase mass balance. In contrast, the distribution in the 3D model is broadened. Nevertheless, compared to a fully segregated laminar flow, where the breakthrough is expected at half of the mean residence time, molecular diffusion in lateral direction contracts the distribu-

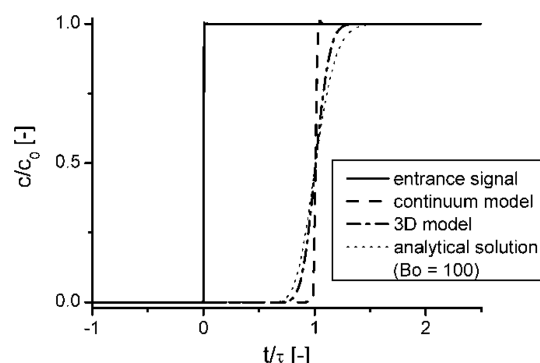


Figure 2. Cumulative residence time distribution.

tion. Under reference conditions, the product $Re \cdot Sc = 30$ is in the region where the residence time distribution is narrowest ($Re \cdot Sc = 13.9$) according to the Aris–Taylor model¹⁵ for laminar flow in circular tubes.

To estimate the influence of the broadened residence time distribution compared to plug flow, a simple axial dispersion model has been employed. It has been found that the residence time distribution is smaller than predicted by the dispersion model at a Bodenstein number of $Bo = 100$ and hence the broadened residence time distribution has hardly any effect on the reactor performance. In fact, the discrepancy concerning the conversion at high Damköhler numbers is caused by a distinct washcoat to void ratio of the one deformed channel compared to the others. Accordingly, averaging of the washcoat fraction employed for the analytical solution and the continuum model lowers the prediction of conversion compared to the 3D model. Under the chosen conditions, however, the differences caused by the residence time distribution and washcoat ratio only slightly influence the model predictions and hence enable for the comparison concerning the heat transfer properties.

3.2. Thermal Resistance Models under Nonreactive and Stagnant Fluid Conditions. Before comparing the models under reactive conditions, it is studied whether the models for the effective radial dispersion coefficient are able to predict the properties of the low cell number monoliths adequately, since their derivation and validation were based on an imposed heat flux perpendicular to an elementary cell comprising a single channel. The boundary conditions are set similar to the approach of Hayes et al.¹¹ for high channel number monolith structures. A constant heat flux is imposed at a coaxial tube completely located in the solid matrix while the temperature of the lateral surface is held constant (Figure 3a). Figure 3b compares the radial temperature profiles of the different models concerning the effective radial dispersion coefficient to the predictions of the 3D model. If plotted as a function of distance to the center, the temperatures of the mesh elements scatter due to the lack of axial symmetry in the 3D model. Good agreement is found for the symmetric model, though the direction of heat flux diagonal to an elementary cell significantly contributes to the heat flux (see heat flux lines in Figure 3a). This is different to the boundary conditions set for an elementary cell at the validation of the symmetric model by Visconti et al.¹³ The series and parallel model under- and overestimate the maximum temperature respectively, which is in line with the findings of Hayes et al.¹¹ for various void fractions concerning an array of channels. It is noteworthy that for calculation of the radial effective properties not the integral values of the void fraction over the whole cross-section (0.60)

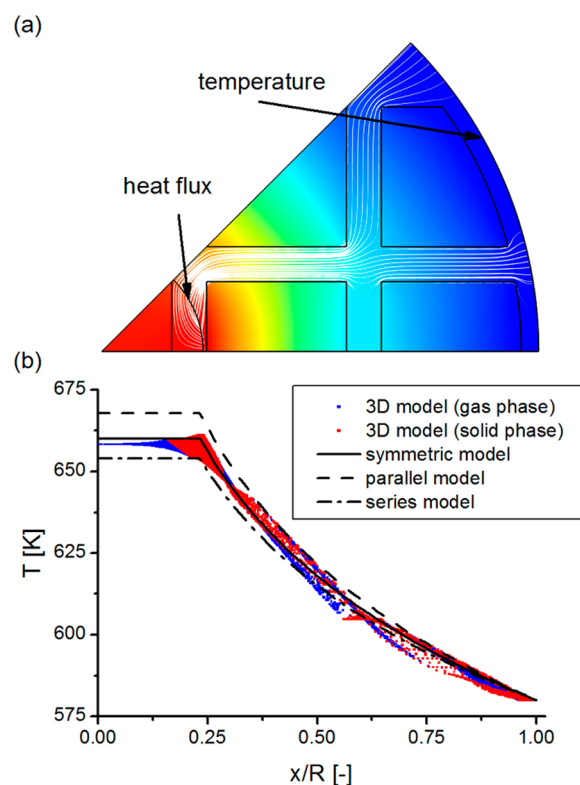


Figure 3. Temperature profiles under nonreactive conditions.

but over a single, nondeformed channel (0.64) need to be employed at this low channel number monolith.

3.3. Comparison at Polytropic Operation. At polytropic operation, sharp temperature gradients between the wall and gas phase call for a finer mesh than at the previous conditions. Figure 4 proves that results for the model comparison are independent of the mesh size. Slight differences are observed mainly for the temperature profile in the middle of the central channel (Figure 4c) concerning the two coarsest meshes. The differences of maximum temperature are less than 1 K concerning all meshes. Owing to low lateral concentration gradients under the reference conditions the concentration profiles vary only slightly even for the coarsest meshes.

Figure 5 compares the two models in terms of the maximum temperature and conversion. An excellent agreement is found for the reference conditions: maximum temperatures differ by less than 0.5 K. In addition, Figure 6 illustrates that the continuum model also predicts axial profiles of the solid phase temperature at various radial positions very well. This still holds when varying the conditions by either changing the activation energy or the feed concentration for moderate Damköhler numbers (Figure 5). At higher reaction rates, however, the models deviate: maximum temperatures and conversions in the 3D model are systematically higher. There are numerous studies in the literature addressing the comparison of 1D, 2D axisymmetric, and 3D single channel models for square ducts under adiabatic conditions.¹⁴ In general, they come to the conclusion that 1D models do not guarantee an adequate description under all operation conditions. For example Groppi et al.¹⁶ compared 1D and 3D single channel models at adiabatic operation for a catalytic combustion and revealed that the 1D model failed in describing the wall temperature profile. This was attributed to the lack in predicting local heat and mass transfer coefficients in the light-off region. Similarly Heck et

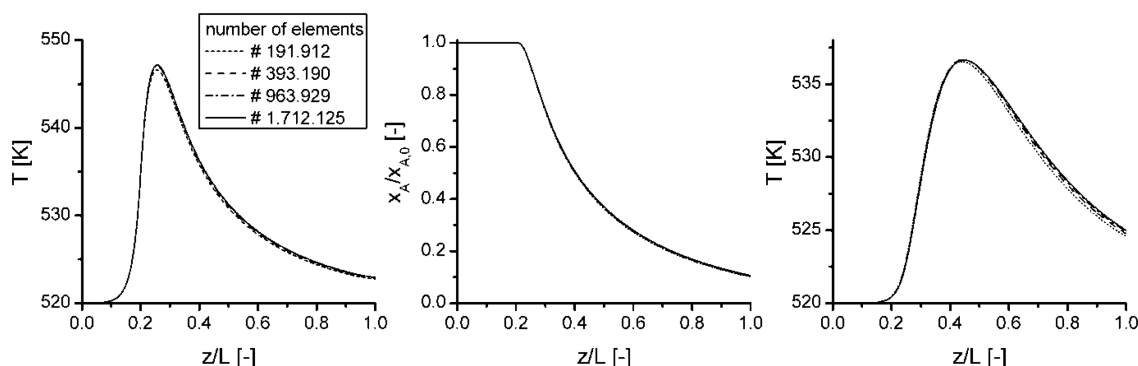


Figure 4. Mesh refinement at reference conditions ($DaI = 2.0$): (a) axial profile of solid phase temperature ($r = 2$ mm), (b) axial profile of normed mole fraction ($r = 0$), (c) axial profile of gas phase temperature ($r = 0$).

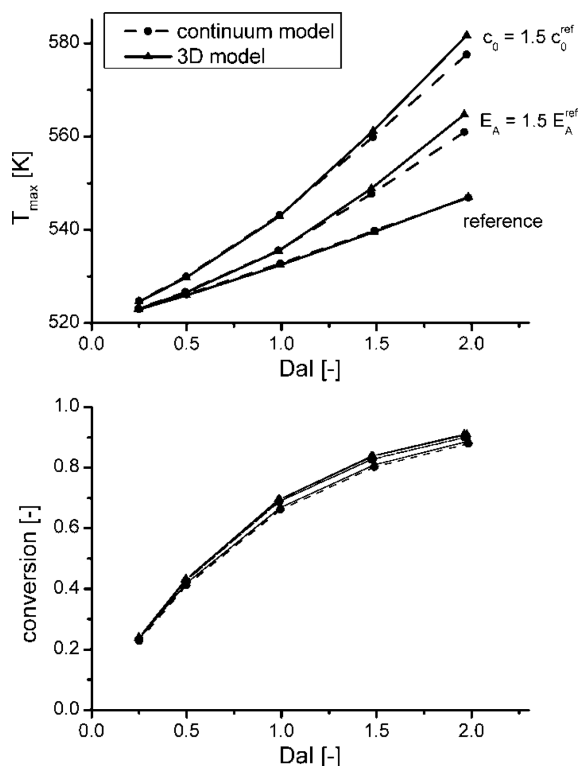


Figure 5. Model comparison at polytropic operation in terms of maximum temperature and conversion.

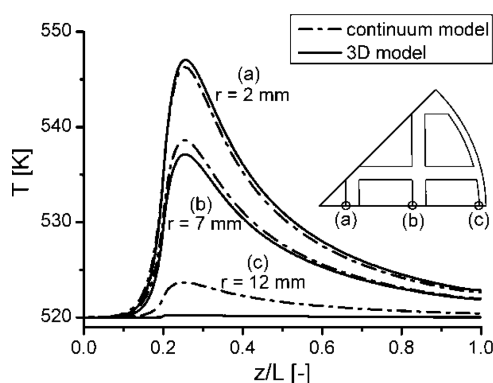


Figure 6. Axial profiles of solid phase temperature at different radial positions at reference conditions ($DaI = 2.0$).

al.¹⁷ found a sharp increase in Nusselt numbers determined from radial temperature gradients using a 2D model in the region of the light-off. However, the performance could be described almost quantitatively using asymptotic Nusselt numbers in a 1D model. Canu and Vecchi¹⁸ observed that even 2D and 3D models predict a different light-off behavior because of a faster heating of corners in a square channel which cannot be described using an axisymmetric 2D model. The continuum model employed in this study also suffers from the 1D channel assumption and the usage of global transfer coefficients. **The correlations for calculation of the Nusselt and Sherwood number are based on the analogy with the Graetz problem with constant wall temperature.**⁴

However, possible inadequacies of heat and mass transfer coefficients are not the only reason for the deviations observed in Figure 5 at high reaction rates. In fact, heat transfer from the solid to the gas phase is only a minor contribution to the total heat removal. If the heat transfer to the gas phase is neglected, the maximum temperature will rise only about 3 K for the fastest reaction rate employed in Figure 5 under reference conditions. We found that the deviations at fast reaction rates are highly sensitive on the diffusion coefficient. The difference is even more pronounced for high diffusion coefficients, where there are hardly any concentration gradients between the wall and channel center in the 3D model and the solid and gas phase in the continuum model correspondingly. If the concentration in the solid phase is set equal to that of the gas phase, the temperature rise in the continuum model will indeed be elevated, but will not reach the maximum temperature in the 3D model under harsh conditions and high diffusion coefficients. This indicates that not only the description of mass transfer via the mass transfer coefficient but the continuum approach itself is the reason for this deviation: in the chosen monolith structure there are only three channels in the radial direction. A hot spot will form at the metallic wall of the inner channel and will locally cause faster reaction rates. Lateral diffusion in the central and the surrounding channels to the hot spot amplifies this effect. In the continuum model, however, it is intrinsically assumed that the channel size is small compared to the tube size. Consequently, the flow is radially completely segregated, and hence diffusion to the position of higher reaction rates is not possible. Nevertheless, just as discrepancies between 1D and more detailed single channel models are observed mainly under harsh conditions near the light-off of highly exothermic reactions, the continuum model predicts results very close to the 3D CFD model at moderate conditions that are more relevant for the industrial scale

production of chemicals via a wide range of reactions like partial oxidations or hydrogenations.

4. CONCLUSIONS

In this work we presented a comparison of a pseudocontinuous, heterogeneous 2D conductive honeycomb model to a detailed 3D CFD model. First, we compared the models under isothermal conditions to the analytical solution of an ideal PFR to ensure correct implementation. As it has been reported in the literature that at adiabatic operation 1D channel models do not suffice for an adequate description under all conditions, we discussed and quantified the influence of the residence time distribution, mass transfer to the washcoat and the washcoat to void ratio of the individual channels on deviations between the models. The comparison under polytropic conditions revealed that the continuum model allows for an excellent description of the reactor performance. The deviations observed at high reaction rates could be attributed to the usage of global heat and mass transfer coefficients derived from the Graetz problem at constant wall temperature and the lack of the continuum model to include mass transfer via lateral diffusion to the hot spot. However, these effects influence the reactor performance only at harsh conditions of large radial temperature gradients that, following the conclusions of Groppi and Tronconi⁴ concerning the optimum design of metallic honeycombs, are to be avoided by designing the honeycombs optimally in terms of structure, void fraction, and material. In general, the continuum approach allows for an adequate description under polytropic conditions at low computational efforts and high flexibility in terms of different structures.

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Notes

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NOMENCLATURE

Formula Symbols

D_A = diffusion coefficient, m^2/s
 DaI = Damköhler number ($= \xi k L / u$)
 E_A = activation energy, J/mol
 ΔH_R = reaction enthalpy, J/mol
 I = identity matrix
 K_h = heat transfer coefficient, $W/(m^2K)$
 K_m = mass transfer coefficient, $kg/(m^2s)$
 L = reactor length, m
 M = molar mass, kg/mol
 T = temperature, K
 W = specific mass flow rate, $kg/(m^2s)$
 W_t = mass flow rate, kg/s
 c = concentration, mol/ m^3
 c_p = heat capacity at constant pressure, J/(kgK)
 d = diameter, m
 k = reaction rate constant, 1/s
 k_0 = preexponential factor, 1/s
 m = pitch size, m
 n = normal vector
 p = pressure, Pa
 r = radial coordinate, m
 t = time, s

u = velocity vector, m/s
 w = mass fraction
 z = axial coordinate, m
 δ = thickness of washcoat, m
 λ = thermal conductivity, $W/(m \cdot K)$
 μ = viscosity, Pa·s
 ξ = volume fraction of washcoat
 ρ = density, kg/m^3
 τ = hydrodynamic residence time, s

Subscripts

A species A
ax axial
e effective
g gas phase
r radial
ref reference conditions
s solid phase
w wall

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